Demonstrating vicarious learning through agentic student-teacher interactions

and Mats Kroneberg and
Author Three and Author Four

Conversational Agents Course – Final Report
Heriot-Watt University, Edinburgh, United Kingdom
{a1, mmk12, a3, a4}@hw.ac.uk

Abstract

[MARKING: This section contributes to the "Abstract, Aims, Objectives and Project Description" criterion (10 marks). The abstract must succinctly describe the project. Cover: (1) the problem/task addressed, (2) your approach and key components, (3) evaluation methods used, and (4) main findings. Keep to 100–200 words. The marker will check that research aims are immediately apparent from reading this.]

1 Introduction

[MARKING: The introduction also contributes to the "Aims, Objectives and Project Description" criterion (10 marks). Make the research aims and objectives of the project unmistakably clear. The marker will look for: (a) a well-motivated problem statement, (b) explicit research aims / objectives, (c) a summary of the approach taken, and (d) a roadmap of the report.]

1.1 Problem Statement and Motivation

1.2 Research Aims and Objectives

[State your aims and objectives explicitly. Consider using a short enumerated list so the marker can see them at a glance.]

1.3 Report Structure

In the following sections, we discuss previous work (Section 2), present the system design and architecture (Section 3), describe the implementation details (Section 4), present our evaluation and results (Section 5), and conclude with recommendations for future work (Section 6).

2 Previous Work

[MARKING: This is the highest-weighted section at 40 marks. The marker will assess: (1) clarity and readability of writing, (2) whether references are appropriate, thorough and properly cited, (3) clear logical organisation, (4) [MSc only] critical analysis of the field — do not just summarise papers but compare, contrast, and identify gaps, and (5) [MSc only] commentary on how the reviewed technologies apply to emerging application areas. Aim for a well-structured narrative, not a list of paper summaries.]

2.1 Background and Context

[Set the scene. Explain foundational concepts the reader needs to understand the rest of the review (e.g. how humans process language, key terminology, the domain your system operates in).]

2.2 Related Systems and Architectures

[Survey the main category of systems your work relates to. Describe the evolution from earlier approaches to current ones. Cite key papers and explain how each contributes to the field.]

2.2.1 Specific Paradigm or Framework

[Drill into the specific paradigm your system builds on (e.g. incremental processing, end-to-end models, retrieval-augmented generation). Explain the framework, its key concepts, advantages and limitations.]

2.2.2 Recent Developments

[Cover the most recent and relevant work. This demonstrates thorough and up-to-date engagement with the literature.]

2.3 Key Techniques and Components

[Review the specific techniques your system uses (e.g. ASR models, LLM-based dialogue, slot filling, intent recogni-

tion). For each, discuss the state of the art and justify why you chose the approach you did. 201 (2011)]

2.4 Critical Analysis and Emerging Applications

[[MSc students: this sub-section is explicitly assessed. 4th-year students will also benefit from including it.] Synthesise the literature — do not just summarise. Compare and contrast the approaches reviewed. What are the open problems? Where do current methods fall short? Additionally, comment on applicability: how do the technologies and methods discussed apply to emerging application areas (e.g. multimodal assistants, healthcare, education, accessibility)?]

2.5 Evaluation Methods in the Literature

[Review how systems like yours are typically evaluated. This motivates and justifies the evaluation approach you will describe later.]

2.5.1 Intrinsic Evaluation

2.5.2 Extrinsic Evaluation

3 System Design and Architecture

[MARKING: This section is worth 10 marks. The marker will check: (1) that each system component is clearly described and its inclusion is well motivated, and (2) that the overall design is coherent — i.e. the components fit together logically. Include an architecture diagram. Explain why you chose this design, not just what *it is.]*

3.1 Architecture Overview

hello 201 (2011)

[Provide a high-level overview of the full system. Reference your architecture diagram (Figure 1). Explain how data flows through the system from input to output.]

3.2 Component A — [Name]

[Describe this component's role in the system, its inputs and outputs, and why it was chosen over alternatives. Keep this at the design level — save low-level implementation details for Section 4.]

3.3 Component B — [Name]

3.4 Component C — [Name]

3.5 Design Decisions and Justifications

[Summarise the key design trade-offs you made and why. This helps demonstrate that the overall design is coherent and well-motivated rather than arbitrary.]

4 Implementation

[MARKING: This section is worth 20 marks. The marker will assess: (1) whether a significant amount of implementation work has been done, and (2) whether the resulting system functions well and is of appropriate complexity. Focus on how *you built the system: technical details, libraries/APIs used, algorithms, data structures, challenges encountered and how you solved them. Include code snippets, pseudocode, or detailed descriptions where they help convey complexity and effort. If you compared alternative implementations, describe the comparison and rationale.]*

4.1 Component A — Implementation Details

[Describe the implementation of this component. What libraries, models, or APIs did you use? What was non-trivial about making it work?]

4.2 Component B — Implementation Details

[Detail the implementation steps. For a dialogue manager, for example, describe the slot-filling loop, prompt engineering, error handling, state management, etc.]

4.3 Component C — Implementation Details

4.4 Integration and Challenges

[Describe how the components were integrated into a working system. Discuss any significant challenges encountered during development and how they were resolved. This demonstrates the depth of work done and helps justify the "significant amount of work" criterion.]

5 Evaluation

[MARKING: This section is worth 20 marks. The marker will check: (1) that evaluation methods and metrics are discussed thoroughly, (2) that you had an appropriate evaluation plan, (3) that the method is clear and sensible, and (4) that results and conclusions are presented clearly. Use tables and figures to present results. Interpret your results — do not just state numbers.]

Briefly introduce your evaluation strategy, the scenario(s) tested, and any datasets used.

5.1 Intrinsic Evaluation

[Describe the intrinsic evaluation: what was measured, what data was used (e.g. dataset, number of test items), and what metrics were applied. Define each metric with a formula if appropriate.]

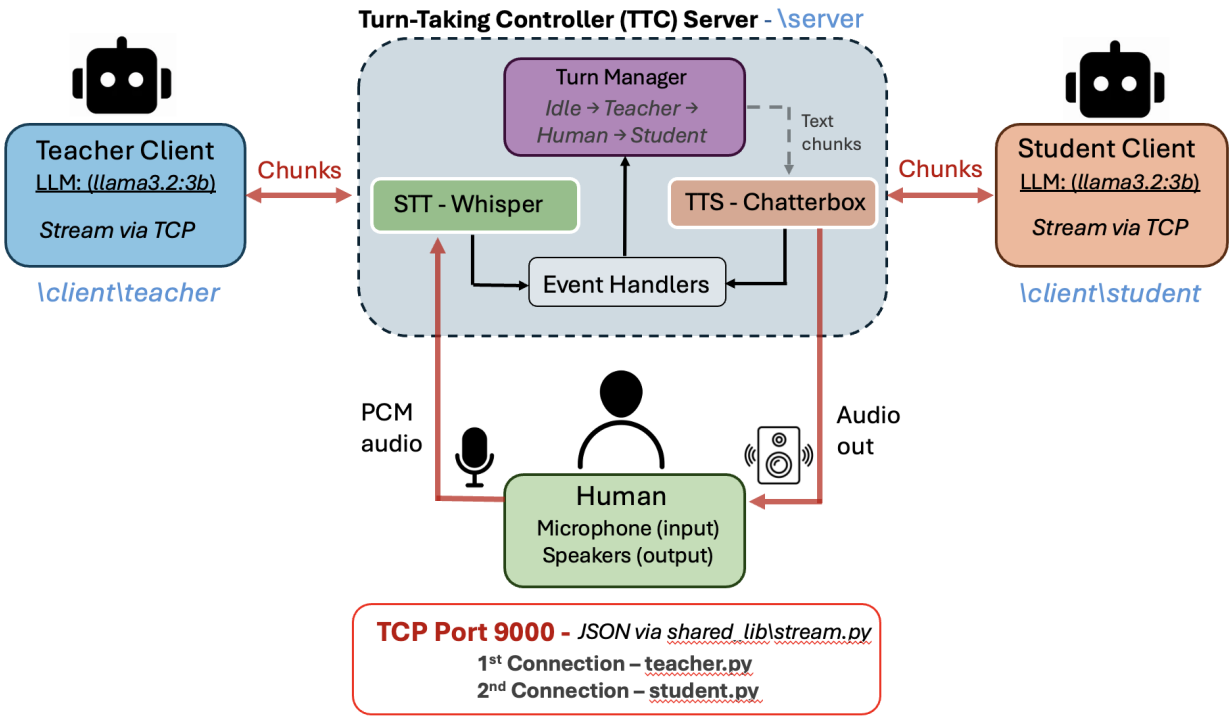

Figure 1: Architecture of the proposed system.

5.1.1 Results

[Present intrinsic results clearly in a table. Interpret them — what do they tell you about your system's performance? Compare with baselines or prior work where possible.]

Metric	Condition A	Condition B
Metric 1	0.000	0.000
Metric 2	0.000	0.000
Metric 3	0.000	0.000

Table 1: Intrinsic evaluation results.

5.2 Extrinsic Evaluation

[Describe your extrinsic evaluation plan: participant recruitment, study procedure, ethical approval, and any measures taken to ensure data quality (e.g. attention checks, exclusion criteria).]

5.2.1 Task Performance

[Report objective task-level results: task success rate, number of turns, slot-filling accuracy, dialogue efficiency, etc.]

5.2.2 Behavioural Analysis

[If applicable, analyse how the system handled specific user behaviours (e.g. pauses, self-corrections, interruptions). Present in a table.]

Behaviour	Occurrences	Success Rate
Behaviour A	0	0%
Behaviour B	0	0%
Behaviour C	0	0%

Table 2: Success rates for handling different behaviours.

5.2.3 Subjective Evaluation

[Report questionnaire results. Describe the instrument (e.g. Godspeed, Likert scales), validity checks (KMO, Cronbach's alpha), and statistical tests used. Present frequencies and averages with figures if possible.]

5.3 Discussion of Results

[Tie all evaluation results together. What do the intrinsic and extrinsic results tell you collectively? Were there any surprising findings? What are the limitations of your evaluation? This is where you demonstrate clear and sensible interpretation — a key part of the marking criterion.]

6 Conclusions

[Summarise the main contributions and findings. Acknowledge limitations honestly. Suggest concrete directions for future research. This section contributes to the overall impression of whether results and conclusions are presented clearly (part of the 20-mark Evaluation criterion).]

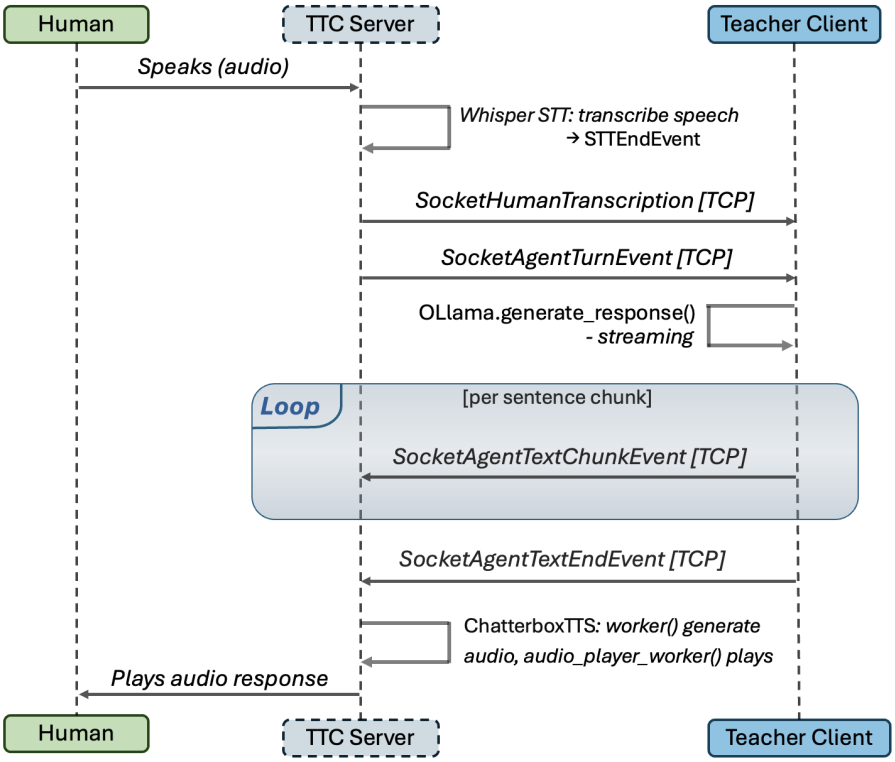

Figure 2: Sequence diagram of the teacher-only interaction workflow (Human, TTC Server, Teacher Client).

[The quality and thoroughness of your references contributes to the Literature Review mark (40 marks). Ensure citations are consistent, complete, and drawn from reputable sources (peer-reviewed papers, established textbooks, official documentation).]

References

(2011). *Cartesian genetic programming / Julian F. Miller, editor.* Natural computing series. Springer, New York, 1st ed. 2011. edition.

A Work Assignment

[Describe each team member's or sub-team's responsibilities. This helps the marker understand individual contributions.]

A.1 Team A: [Names]

A.2 Team B: [Names]

A.3 Team C: [Names]

B Questionnaire

[Include the full questionnaire used in extrinsic evaluation so the marker can assess its appropriateness.]

Aspect	Description	Score
Aspect 1	Low (1) – High (5)	
Aspect 2	Low (1) – High (5)	
Aspect 3	Low (1) – High (5)	

Table 3: The questionnaire applied in this report.

C Supplementary Results

[Include any additional tables, statistical tests (e.g. Wilcoxon), or detailed breakdowns that support but would clutter the main text.]

D Ethical Considerations

[Address: (1) informed consent, (2) data protection and GDPR compliance, (3) participant rights (voluntary participation, right to withdraw), (4) risk mitigation, and (5) broader considerations such as bias in conversational agents.]

E Consent Form

F UI Screenshots

[Include screenshots showing typical system behaviours during interaction (e.g. session start, slot filling, handling pauses, self-correction, interruption, session end). These help demonstrate that the system functions well (Implementation criterion, 20 marks).]

Marking Criteria Summary — DELETE BEFORE SUBMISSION

Criterion	Marks	Report Sections
Abstract, Aims, Objectives & Project Description – Abstract succinctly describes project – Research aims are clear	**10**	Abstract, §1
Literature Review – Clarity / readability – Appropriate / thorough references – Good and clear organisation – Critical analysis (MSc only) – Applicability commentary (MSc only)	**40**	§2
System Design and Architecture – Components clear and well motivated – Overall design coherent	**10**	§3
Implementation – Significant amount of work done – System functions well, appropriate complexity	**20**	§4
Evaluation – Methods and metrics discussed thoroughly – Appropriate evaluation plan – Method clear and sensible – Results / conclusions presented clearly	**20**	§5, §6
Total	**100**	

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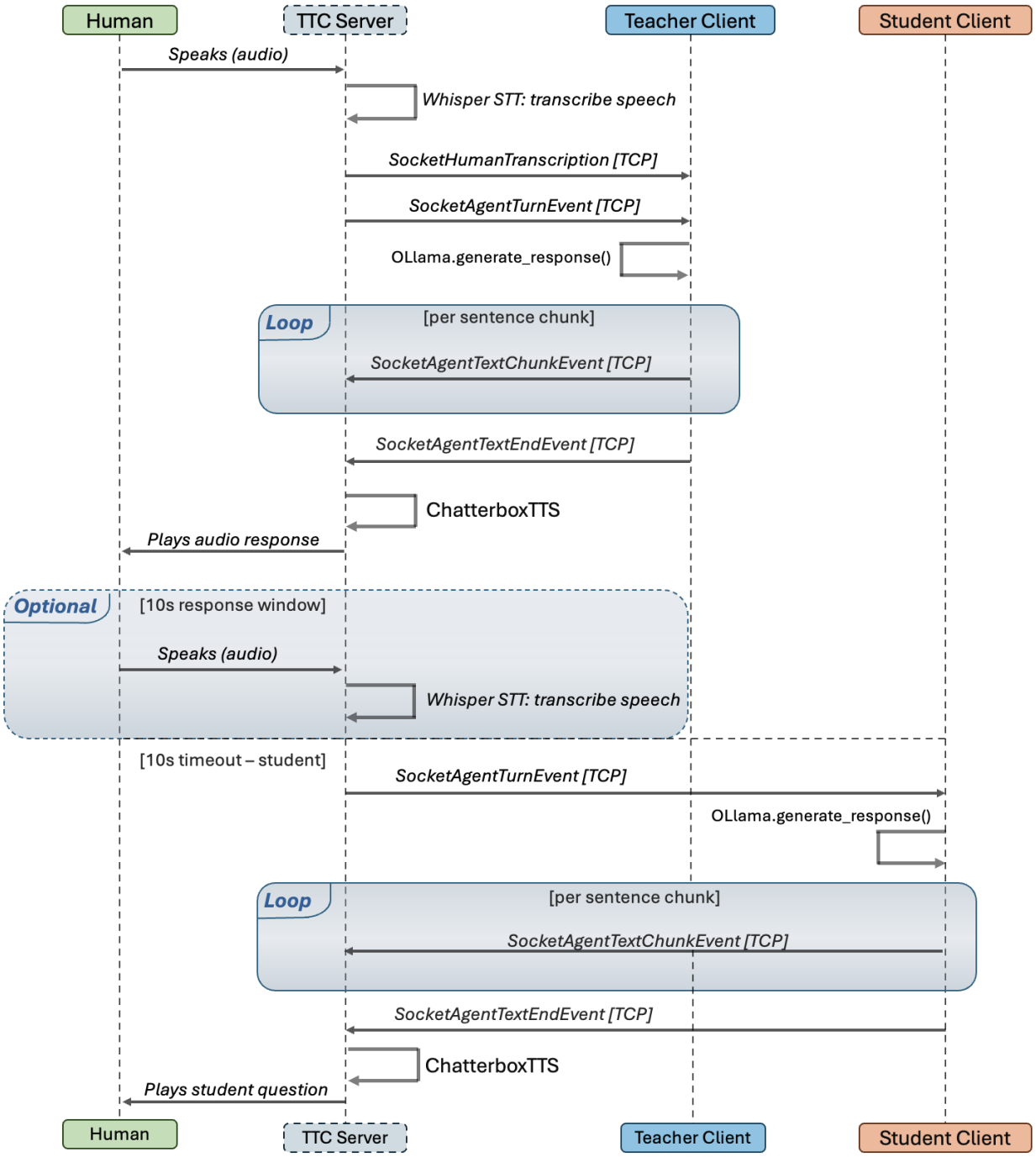

Figure 3: Sequence diagram of the combined teacher–student interaction workflow (Human, TTC Server, Teacher Client, Student Client).